

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – IV
PAPER –1
TEST DATE: 31-07-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. C

Sol. Potential of centre of sphere = $\frac{Kq}{r} + V_i = \frac{Kq}{r}$

where V_i = potential due to induced charge at centre = 0 [$\therefore \Sigma q_i = 0$ and all induced charges are equidistance from centre]

$\therefore$ potential at point $P = \frac{Kq}{r} = \frac{Kq}{r_1} + V_i$ (For conductor all points are equipotential)

$$\therefore V_i = K \left(\frac{q}{r} - \frac{q}{r_1} \right)$$

2. A

Sol. Applying Snell's law between the points O and P , we have

$$2 \times \sin 60^\circ = (\sin 90^\circ) \times \frac{2}{(1+H^2)}, \quad 2 \times \frac{\sqrt{3}}{2} = 1 \times \frac{2}{(1+H^2)}$$

$$(1+H^2) = \frac{2}{\sqrt{3}}, \quad H = \sqrt{\left(\frac{2}{\sqrt{3}} - 1 \right)}$$

3. D

Sol. Impulse on ball

$$I = (1+0.5)\sqrt{2gh} \text{ m (vertically)}$$

Frictional impulse on block (horizontally)

$$= \mu I = 0.2 \times 1.5 \text{ m} \sqrt{2gh}$$

Decrease in velocity

$$\Delta V = \frac{\mu l}{m} = 0.3 \sqrt{2gh}$$

4. A

Sol. From first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = nC_v (T_2 - T_1) + \frac{nR(T_1 - T_2)}{x-1}$$

$$\frac{1}{n} \left(\frac{\Delta Q}{\Delta T} \right) = C_v - \frac{R}{x-1}$$

$$\therefore (C - C_v)(x-1) = -R$$

$\therefore$ graph is a rectangular, hyperbola

$\therefore$ co-ordinate of $P_1 (0, C_p)$

i.e. $\left(0, \frac{5}{2}R\right)$ and that of $P_4 \left(\frac{5}{3}, 0\right)$

5. CD

Sol. If $\mu_1 = 0, \mu_2 \neq 0$

$t_1 = 0$, balancing torques about A

$$N_1 \ell \sin \theta = mg \frac{\cos \theta}{2}$$

$$N_1 \tan \theta = \frac{mg}{2}$$

If $\mu_1 \neq 0, \mu_2 \neq 0$

$T_2 = 0$, equilibrium cannot be attained.

If $\mu_1 \neq 0, \mu_2 \neq 0$

$$N_1 = t_2 = \mu_2 N_2 \quad ; \quad N_2 + t_1 = mg$$

$$N_2 + \mu_1 N_1 = mg \quad ; \quad N_2 + \mu_1 (\mu_2 N_2) = mg \quad ; \quad N_2 = \frac{mg}{1 + \mu_1 \mu_2}$$

6. ABCD

Sol. Let elongation in spring A, B and C be x_1, x_2 and x_3 respectively.

Considering spring forces and constraint relations

$$x_2 = 4x_3 \quad \dots (i)$$

$$x_2 = 2x_1 \quad \dots (ii)$$

$$\text{and } x_1 + 2x_2 + x_3 = x \quad \dots (iii)$$

$$\Rightarrow x_1 = \left(\frac{2}{11}\right)x \quad ; \quad x_2 = \left(\frac{4}{11}\right)x \quad ; \quad x_3 = \left(\frac{1}{11}\right)x$$

$$\text{Also, } F = 2K \left(\frac{x}{11}\right)$$

$$\Rightarrow T = 2\pi \sqrt{\frac{11m}{2k}}$$

7. ABC

Sol. $\Delta U = \frac{fR\Delta T}{2}$

$\Delta W = \frac{nR\Delta T}{1-x}$ where $PV^x = \text{constant}$. Here $x = -\frac{1}{2}$

8. ACD

Sol. The gravitational force due to these masses on a mass at O is equal and opposite.

So, the resultant force is zero hence, the resultant field is zero

Also, any point on y-z plane which is equidistant from two cavities will have zero field intensity hence constant potential.

9. ABCD

Sol. $x + \alpha = \beta + x = 90^\circ \Rightarrow \alpha = \beta$

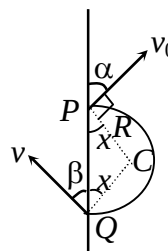
Also, $PQ = 2R \cos(90 - \alpha) = 2R \sin \alpha = \frac{2mv_0}{qB} \sin \alpha$

$|\vec{v}| = \text{constant}$

$\therefore v = v_0$

For 2π rotation it takes $\frac{2\pi m}{qB}$ time

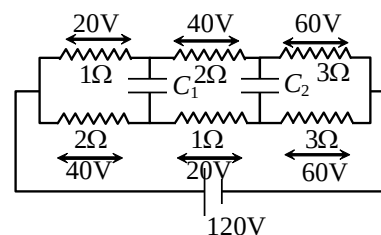
For $(2\pi - 2\alpha)$ it takes $\frac{2m(\pi - \alpha)}{qB}$



10. ACD

Sol. Now, potential difference across C_1 is 20 V and across C_2 is zero.

$\therefore$ charge stored in C_1 is 40 μC and in C_2 is zero.



Section - B

11. 6

Sol. $\frac{\Delta Q}{\Delta t} = K_A A \frac{dT}{dx}$

$\frac{dT}{dx}$ in conductor A = slope of graph = $\sqrt{3}$

Since both conductors are connected in series, same heat current will flow in A and B.

$K_A (dT/dx)_A = K_B (dT/dx)_B$

$K_B = 6$

12. 7

Sol. Given, voltage $V = (100 \pm 5)\text{V}$

Current $I = (10 \pm 0.2)\text{A}$

According to Ohm's law, $V = IR$ or $R = V/I$

Taking log of both sides,

$$\log R = \log V - \log I$$

Differentiating, we get

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} - \frac{\Delta I}{I}$$

$$\text{For maximum error, } \frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

Multiplying both sides by 100 for taking percentage, we get

$$\frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 100 + \frac{\Delta I}{I} \times 100$$

Percentage error in resistance R

$$\begin{aligned} &= \frac{\Delta V}{V} \times 100 + \frac{\Delta I}{I} \times 100 \\ &= \frac{5}{100} \times 100 + \frac{0.2}{10} \times 100 = 7\%. \end{aligned}$$

13. 2

Sol. For getting null point

$$\frac{R_1}{R_3} = \frac{R_2}{R_4}$$

as $R_2 = R_4$

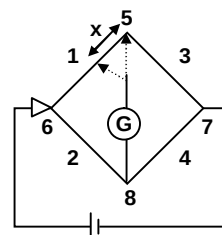
$$R_1 = R_3$$

Let the pointer at point 5 is moved to left by distance x to get null point as shown in the figure. If resistance per unit length of wire 3 is r then that of wire 1 will be $2r$.

$$(8 - x)2r = x \times 2r + 8 \times r$$

$$4x = 8$$

$$x = 2\text{m}$$



Section – C

14. 0.80

$$\text{Sol. } \Delta Q = (10 \text{ g}) (80 \text{ cal/gm}) = 800 \text{ cal.}$$

15. 82.50

$$\begin{aligned} \text{Sol. } \Delta Q &= (1) (0.5) (5) + (1) (80) \\ &= 82.5 \text{ cal} \end{aligned}$$

16. 3.18

17. 0.05

Sol. (Combined solution 18 & 19)

$$V = \frac{0.5 \times 3}{(0.5 + 1)} = 1 \text{ m/s} \Rightarrow K_{\max} = U_{\max}$$

$$\frac{1}{2} \times 1.5 \times 1^2 = \frac{1}{2} \times 600 \times A^2$$

$$A = 5 \text{ cm}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{600}{1.5}} = \frac{10}{\pi}$$
$$f = \frac{10}{\pi} \text{ Hz}$$

18. 0.25

19. 0.20

Sol. (Combined solution Q.18 & 19)

General equation of wave is $y = A \sin(\omega t - kx + \phi)$.

Chemistry

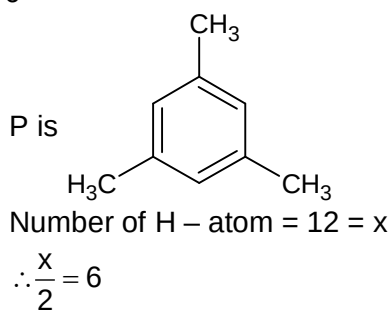
PART – II

Section – A

20. D
Sol. Hybridization of boron changes to sp^3 .
21. C
Sol. Both NaCl and KNO_3 are neutral salt.
22. C
Sol. KH is not formed as there is no H_2 in the atmosphere.
23. A
Sol. The products are CH_3OH , CH_3I , C_2H_5OH and C_2H_5I .
24. ACD
Sol. The ratio of [salt] to [acid] or [base] should be within 1 : 10 or 10 : 1 for buffer solution.
25. BCD
Sol. $CH_3CH = CH_2$ will form alcohol upon hydrolysis.
26. BCD
Sol. Silicones contain carbon, silicone and oxygen. Silanes and selenes contain silicone and hydrogen.
27. ACD
Sol. These complexes contain complex ion cation and anions.
28. ABC
Sol. The precipitate is $Fe_4[Fe(CN)_6]_3$.
29. AB
Sol. $E_{\text{Cell}} = E_{\text{Cell}}^0$ if $\frac{C_1}{(C_2)^2} = 1$

Section – B

30. 6
Sol.



31. 8

Sol.
$$\text{pH} = \frac{1}{2}(\text{p}^{K_{a_1}} + \text{p}^{K_{a_2}}) = \frac{1}{2}(6 + 10) = 8$$

32. 14

Sol. A = SO₂, B = SO₃, C = H₂S₂O₇
H₂S₂O₇ contains 14 covalent bonds

Section – C

33. 1.00

34. 0.06
(range 0.059 to 0.06)

Sol. (Q.33 & 34):

For $E_{\text{cell}} = E_{\text{cell}}^0$, the value of C is 0.1

$$\therefore \text{pH} = -\log 0.1 = 1$$

$$E_{\text{H}^+/\text{H}_2} = E_{\text{H}^+/\text{H}_2}^0 - \frac{0.0591}{1} \log \frac{1}{[\text{H}^+]}$$

$$= 0 - \frac{0.0591}{1} \log 10 = -0.0591$$

$$\therefore |y| = |-0.0591| = 0.0591 = 0.06$$

35. 3.00

36. 8.00

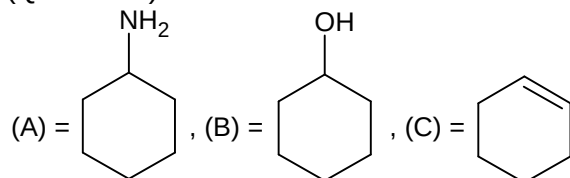
Sol. (Q.35 & 36):

A = P₄O₆, B = H₃PO₃, C = H₃PO₄, D = PH₃

37. 4.00

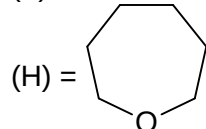
38. 12.00

Sol. (Q.37 & 38):



(D) = HOOC – (CH₂)₄ – COOH, (E) = HOH₂C – (CH₂)₄ – CH₂OH

(F) = HOH₂C – (CH₂)₄ – CH₂Cl, G = O[–]CH₂ – (CH₂)₄ – CH₂Cl



Mathematics

PART – III

Section – A

39. B

Sol. If $f'(x) > 0$ and $f''(x) > 0 \Rightarrow \frac{d^2 f^{-1}(x)}{dx^2} < 0$

If $f'(x) < 0$ and $f''(x) < 0 \Rightarrow \frac{d^2 f^{-1}(x)}{dx^2} < 0$

40. C

Sol. We have, $f(\alpha) = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \left(\frac{\tan x}{\tan x + \cot x} \right) dx \quad \dots(1)$

$\Rightarrow f(\alpha) = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \frac{\cot x}{\cot x + \tan x} dx \quad \dots(2)$

$\therefore$ Adding (1) and (2), we get

$$f(\alpha) = \frac{1}{2}(\cot^{-1}\alpha - \tan^{-1}\alpha) = \frac{1}{2}\left(\frac{\pi}{2} - 2\tan^{-1}\alpha\right) = \frac{\pi}{4} - \tan^{-1}\alpha$$

Clearly, range of $f(\alpha)$ is $\left(-\frac{\pi}{4}, \frac{3\pi}{4}\right)$.

41. B

Sol. $z^5 - 1 = (z - 1)(z - \alpha_1)(z - \alpha_2)(z - \alpha_3)(z - \alpha_4)$

$$\ln(z^5 - 1) = \ln(z - 1) + \ln(z - \alpha_1) + \ln(z - \alpha_2) + \ln(z - \alpha_3) + \ln(z - \alpha_4)$$

Differentiating

$$\frac{5z^4}{z^5 - 1} = \frac{1}{z - 1} + \frac{1}{z - \alpha_1} + \frac{1}{z - \alpha_2} + \frac{1}{z - \alpha_3} + \frac{1}{z - \alpha_4}$$

$$\text{Setting } z = 2, \sum_{r=1}^4 \frac{1}{2 - \alpha_r} = \frac{80}{31} - 1 = \frac{49}{31}$$

42. B

Sol. $\alpha + \beta + \gamma = 6$

$$\alpha\beta + \beta\gamma + \alpha\gamma = 5$$

$$\alpha\beta\gamma = 1$$

$$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = 26$$

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 = 13$$

43. CD

Sol. f_n = number of subset in which n appears + number of subset in which n does not appears.

$S = \{1, 2, 3, \dots, (n-2), (n-1), n\}$ when n appears obviously $(n-1)$ will not appear and when n does not appear up to $(n-1)$ will appear

$$\begin{aligned} \Rightarrow f_n &= f_{n-2} + f_{n-1} \\ f_1 &= 2 \\ f_2 &= 3 \\ \Rightarrow f_3 &= f_1 + f_2 = 2 + 3 = 5 \\ f_4 &= f_2 + f_3 = 3 + 5 = 8 \end{aligned}$$

44. ABCD

Sol. If we could show that $f'''(x) > 0$ then all choices follow since $f(0) = f'(0) = f''(0) = 0$

Indeed $f'(x) = 1 + \frac{8}{3}x^2 - \sec^2 x$, $f'(0) = 0$

$$f''(x) = \frac{16}{3}x - 2\sec^2 x \tan x \quad f''(0) = 0$$

$$f'''(x) = \frac{16}{3} - 2\sec^4 x - 4\sec^2 x \tan^2 x$$

$$= \frac{16}{3} - 2\sec^2 x (\sec^2 x + 2\tan^2 x)$$

$$\because 0 < x < \frac{\pi}{6}$$

$$\max [2\sec^2 x (\sec^2 x + 2\tan^2 x)]$$

$$= 2 \left(\frac{2}{\sqrt{3}} \right)^2 \left(\frac{4}{3} + \frac{2}{3} \right) = \frac{16}{3}$$

$$\Rightarrow f'''(x) > 0$$

Whence all choices follow.

45. ABCD

Sol. Put $x = \frac{1}{x}$ in both the quadratic equation

$$x^2 - 4x + a = 0$$

$$\begin{array}{cc} & \swarrow \quad \searrow \\ \frac{1}{x_1} & \quad \quad \quad \frac{1}{x_3} \end{array}$$

$$x^2 - 6x + b = 0$$

$$\begin{array}{cc} & \swarrow \quad \searrow \\ \frac{1}{x_2} & \quad \quad \quad \frac{1}{x_4} \end{array}$$

$$\frac{1}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}, \frac{1}{x_4} \text{ are in A.P.}$$

$$\text{Let } \frac{1}{x_1} = A, \frac{1}{x_2} = A + D, \frac{1}{x_3} = A + 2D, \frac{1}{x_4} = A + 3D$$

$$\therefore \frac{1}{x_1} + \frac{1}{x_3} = 2A + 2D = 4 \quad \Rightarrow \quad A + D = 2 \quad \dots\dots\dots(1)$$

$$\therefore \frac{1}{x_2} + \frac{1}{x_4} = 2A + 4D = 6 \quad \Rightarrow \quad A + 2D = 3 \quad \dots\dots\dots(2)$$

From, (1) and (2),

$$A = 1, D = 1$$

$$\therefore x_1 = 1, x_2 = \frac{1}{2}, x_3 = \frac{1}{3} \text{ and } x_4 = \frac{1}{4}$$

$$\therefore a = \frac{1}{x_1} \times \frac{1}{x_3} = \frac{1}{3}, b = \frac{1}{x_2} \times \frac{1}{x_4} = \frac{1}{8}$$

Now verify alternatives.

46. ACD

Sol. On expanding we get $(\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{a})(\vec{c}) = -\vec{b}$

$$\Rightarrow \vec{c} - (\vec{a} \cdot \vec{c})\vec{a} = \vec{b}$$

$\Rightarrow$ (a) is correct and (b) is ruled out.

$$\Rightarrow |\vec{c} - (\vec{a} \cdot \vec{c})\vec{a}|^2 = |\vec{b}|^2$$

If θ is the angle between $\vec{a}$ and $\vec{c}$ then

$$4 + (2\cos\theta)^2 - 2(2\cos\theta)^2 = 1$$

$$\Rightarrow \cos\theta = \pm \frac{\sqrt{3}}{2} \quad \Rightarrow \theta = 30^\circ; 150^\circ$$

$\Rightarrow$ (C) and (D) is correct.

47. AB

Sol. $\det(A + B) = -\det(A)\det(B)\det(A + B)$

$$= -\det(A^T)\det(B^T)\det(A + B)$$

$$= -\det(A^T(A + B)B^T)$$

$$= -\det(A^T(A + B)B^T)$$

$$= -\det(A^T A B^T + A^T B B^T)$$

$$= -\det(B^T + A^T)$$

$$= -\det(A + B)^T$$

$$= -\det(A + B)$$

$$\Rightarrow \det(A + B) = 0$$

48. AB

Sol. $\log_3 |2z + 1| < \log_3 |2z - 1|$ (as $\cot^{-1} x$ is a decreasing function)

$$\begin{aligned}
&\therefore |2z+1| < |2z-1| \\
&\text{Or } (2z+1)(2\bar{z}+1) < (2z-1)(2\bar{z}-1) \\
&2(z+\bar{z}) < -2(z+\bar{z}) \\
&z+\bar{z} < 0 \\
&2\operatorname{Re}(z) < 0 \\
&\therefore \operatorname{Re}(z) < 0
\end{aligned}$$

Section – B

49. 1

Sol. In the 1st case,

$$P(\text{they match}) = \frac{{}^3C_2 + {}^nC_2}{{}^{n+3}C_2} = \frac{1}{2}; \frac{6+n(n-1)}{(n+3)(n+2)} = \frac{1}{2}$$

$$\Rightarrow 2(n^2 - n + 6) = n^2 + 5n + 6$$

$$\Rightarrow n^2 - 7n + 6 = 0$$

$$\Rightarrow n = 1 \text{ or } 6$$

In the 2nd case,

$$\frac{3}{n+3} \times \frac{3}{n+3} + \frac{n}{n+3} \times \frac{n}{n+3} = \frac{5}{8}$$

$$\text{Solving, } n^2 - 10n + 9 = 0$$

$$n = 9 \text{ or } 1 \quad \dots\dots\dots(ii)$$

From equation (i) and (ii), $n = 1$

50. 6

Sol. For consistency,

$$\begin{vmatrix} a & 1 & b \\ b & 1 & a \\ a & b & ab \end{vmatrix} = 0$$

$$\Rightarrow (a-b)(a-b^2) = 0$$

$$\Rightarrow \text{Either } a = b \text{ or } a = b^2$$

When $a = b$, then ordered pairs are $(0,0), (1,1), (2,2), (3,3), (4,4)$ When $a = b^2$, then $(4, 2)$

Hence, number of ordered pairs are 6.

51. 1

$$\text{Sol. } \lim_{t \rightarrow x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1$$

Using LH rule

$$\lim_{t \rightarrow x+1} \frac{2t f(x+1) - (x+1)^2 f'(t)}{f'(t)} = 1$$

$$\frac{2(x+1)f(x+1) - (x+1)^2 f'(x+1)}{f'(x+1)} = 1$$

$$2xf(x) - x^2f'(x) = f'(x)$$

$$2xy = (1+x^2) \frac{dy}{dx}$$

$$\frac{2x}{1+x^2} dx = \frac{dy}{y}$$

$$\ln(1+x^2) = \ln|y|$$

$$1+x^2 = cy$$

$$f(0) = 1 \Rightarrow c = 1$$

$$\Rightarrow y = 1+x^2$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{\ln(1+x^2) - \ln 2}{x-1} = \lim_{x \rightarrow 1} \frac{2x}{1+x^2} = 1$$

Section - C

52. 2.00

53. 4.00

Sol. (Q.52 & 53):

a, b, c are three roots of the quadratic equation

$$((4f(-1) - 3)x^2 + (4f(1) - 3))x + f(2) = 0$$

$$\Rightarrow 4f(-1) = 3, 4f(1) = 3 \text{ and } f(2) = 0$$

$$\text{Let } f(x) = (x-2)(Ax+B)$$

$$\text{Now, } 3 = 4f(-1) = 4(-3)(-A+B) \Rightarrow A-B = \frac{1}{4}$$

$$3 = 4f(1) = 4(-1)(A+B) \Rightarrow A+B = \frac{-3}{4}$$

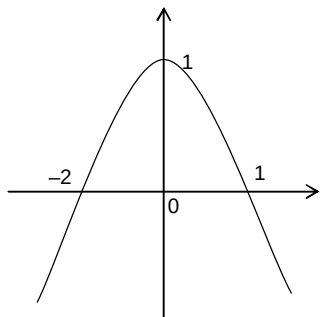
$$\therefore A = \frac{-1}{4}, B = \frac{-2}{4}$$

$$\text{Thus, } f(x) = \frac{1}{4}(4-x^2).$$

The graph of $y = f(x)$ is given in Figure

(A) x coordinates of points of intersection of $y = f(x)$ when the x - axis are ± 2 .

$$(B) \text{ Area} = \frac{3}{2} \int_{-2}^2 \frac{1}{4}(4-x^2) dx = \frac{3}{8}(2) \left(4x - \frac{x^3}{3} \right) \Big|_0^2 = 4$$



54. 4.50

Sol. Equation of the plane π is $A(x-1) + B(y-1) + C(z-1) = 0$ where $A + 0 \cdot B - 1 \cdot C = 0$ and $-A + B + 0 \cdot C = 0$

$\Rightarrow A = B = C$ and the equation of π is $x + y + z = 3$. which meets the co-ordinates axes at A (3, 0, 0), B (0, 3, 0) and C (0, 0, 3). So volume of the tetrahedron.

$$OABC = \frac{1}{6} \begin{vmatrix} 0 & 0 & 0 & 1 \\ 3 & 0 & 0 & 1 \\ 0 & 3 & 0 & 1 \\ 0 & 0 & 3 & 1 \end{vmatrix} = \frac{1}{6} \times 27 = \frac{9}{2}$$

55. 4.50

Sol. Required distance is $\left| \frac{\frac{3\sqrt{3}}{2} + 3\sqrt{3} + 3 - 3}{\sqrt{1+1+1}} \right| = \frac{9}{2}$

56. 5.00

57. 0.50

Sol. (Q.56 – 57):

$f(x) = axe^{-bx}$ has a local maximum at the point (2, 10)

$$\therefore f(2) = 10$$

$$\Rightarrow 2ae^{-2b} = 10$$

$$\Rightarrow ae^{-2b} = 5 \quad (i)$$

$$\text{Now, } f'(x) = a[e^{-bx} - bx e^{-bx}]$$

$$f'(2) = 0$$

$$\Rightarrow a(e^{-2b} - 2be^{-2b}) = 0$$

$$\Rightarrow ae^{-2b}(1 - 2b) = 0$$

$$\Rightarrow b = \frac{1}{2}$$

Putting $b = \frac{1}{2}$ in (i), we get

$$a = 5e$$

$$\therefore a = 5e \text{ and } b = \frac{1}{2}$$